

# HAT-P-54b

R-1879

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_HATP-54\_160125 · created 2026-08-03T11:36:27+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                |
|---------------------------------------------------|--------------------------------|
| Mid-Transit Time (Tmid)                           | 2457412.6624 +/- 0.001 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.147 +/- 0.022                |
| Transit depth (Rp/Rs)^2                           | 2.15 +/- 0.65 [%]              |
| Orbital Inclination (inc)                         | 89.74 +/- 1.41                 |
| Airmass coefficient 1 (a1)                        | 0.964 +/- 0.023                |
| Airmass coefficient 2 (a2)                        | 0.03 +/- 0.017                 |
| Scatter in the residuals of the lightcurve fit is | 2.41 %                         |
| Best Comparison Star                              | #1 - [228, 25]                 |
| Optimal Method                                    | PSF photometry                 |
| Transit Duration (day)                            | 0.0717 +/- 0.0022              |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R* fit vs published    | 0.147 ± 0.022 vs 0.1572 (deviation 0.46σ) |
| Duration fit vs published | 0.0717 d vs 0.075 d (deviation 1.47σ)     |
| Mid-time O–C              | 0.7 min                                   |
| Depth SNR                 | 6.6                                       |
| Residual scatter          | 2.41 %                                    |

## Light curve

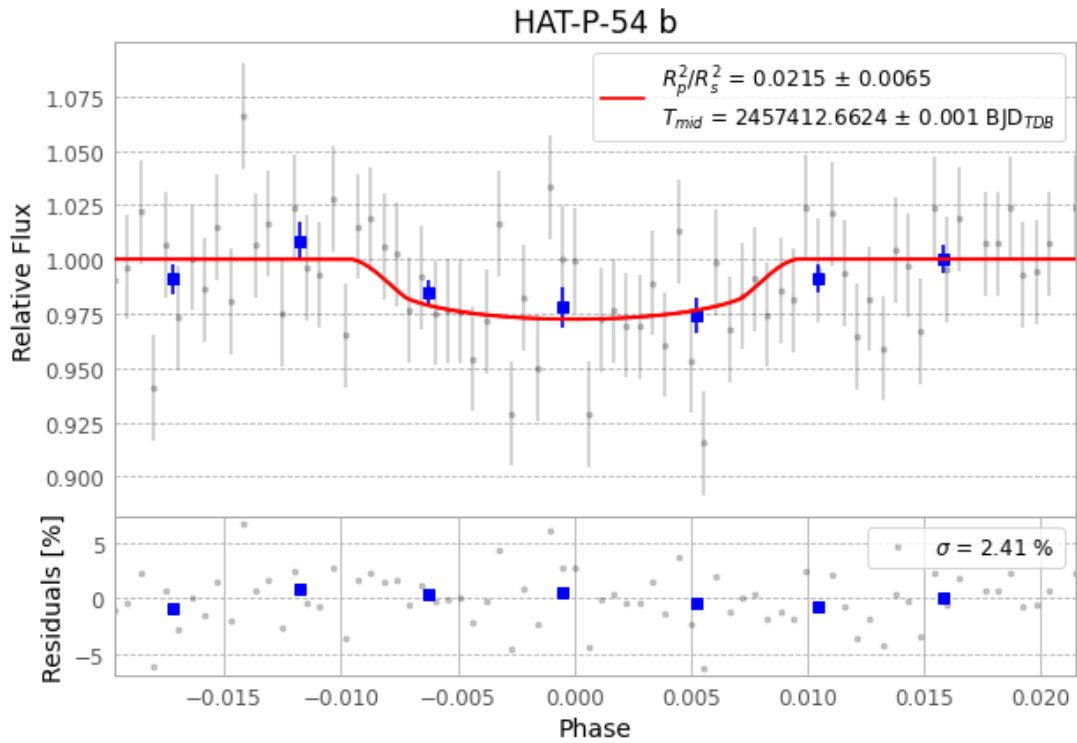


Figure 1 — FinalLightCurve\_HAT-P-54 b\_2016-01-24.png (SHA-256 2f2409d31f072f9d...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [367, 186], 1 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 a5f7cd554e96d904...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit ba3ef8b

## Data provenance

76 FITS frames (50 MB), HATP-54160125015712.FITS → HATP-54160125054212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-160125.log (9c4f9c6afdd3...), FinalLightCurve\_HAT-P-54 b\_2016-01-24.png (2f2409d31f07...), FinalLightCurve\_HAT-P-54 b\_2016-01-24.csv (4ff2ff2f1fa9...)

## Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 11:36Z.